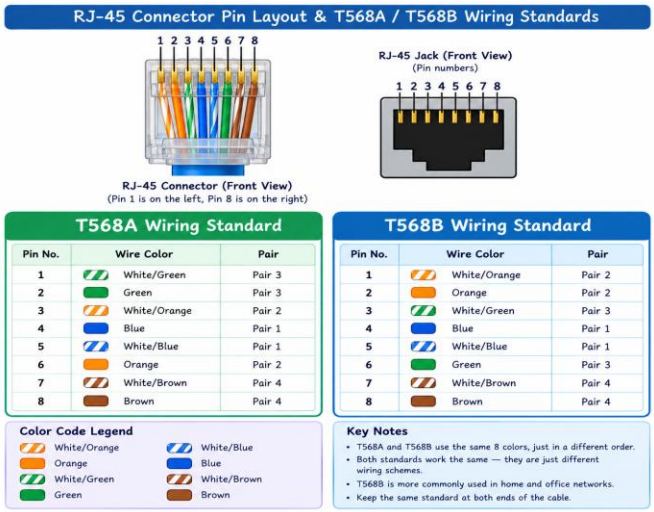
ETHERNET STANDARD AND CONNECTORS

1.Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.

ANS:



2.Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

ANS:

Comparison of Ethernet Standards

Specification	**10BASE-T**	**100BASE-TX**	**1000BASE-T**
Speed	10 Mbps	100 Mbps	1000 Mbps (1 Gbps)
Cable Type	Cat3 or higher UTP	Cat5 or higher UTP	Cat5 or higher UTP
Maximum Cable Length	100 metres	100 metres	100 metres
Connector Type	RJ-45 (8-pin)	RJ-45 (8-pin)	RJ-45 (8-pin)
Common Name	Ethernet	Fast Ethernet	Gigabit Ethernet

**3.Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why.

Hint: Think about how transmit and receive pairs are wired in each cable type.**

ANS:

Connecting Two Laptops Directly Using Ethernet

Cable Type: Crossover Ethernet Cable.

Reason: A crossover cable is traditionally used to connect two similar devices, such as two laptops, directly without a switch or router. It swaps the transmit and receive wire pairs so that one laptop's transmit signals reach the other laptop's receive pins.

- Straight-through cable: Usually connects different types of devices, such as a laptop to a switch or router.
- Crossover cable: Traditionally connects similar devices, such as laptop to laptop.

4.List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

ANS:

Three Devices That Use Ethernet Connections

No.	Device	Likely Ethernet Standard	Use
1	Wi-Fi Router	1000BASE-T	Connects to a computer or switch for high-speed internet.
2	Desktop Computer	1000BASE-T	Provides a fast and stable wired internet connection.
3	Network Printer	100BASE-TX	Connects to the college network for sharing printing services.